

Hybrid CNN-LSTM Model for DDoS Detection and Mitigation in Software-Defined Networks

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Abstract—Software-Defined Networking (SDN) enhances programmability and control but remains highly vulnerable to distributed denial-of-service (DDoS) attacks. Existing solutions often adapt conventional methods without leveraging SDN's native features or addressing real-time mitigation. This study introduces a novel hybrid deep learning framework for DDoS detection and mitigation in SDN, significantly advancing the state of the art. We develop a custom dataset in a Mininet-Ryu testbed that reflects realistic SDN traffic conditions, and employ a multistage feature selection pipeline to reduce redundancy and highlight the most discriminative flow attributes. A hybrid Convolutional Neural Network–Long Short-Term Memory (CNN-LSTM) model is then applied, capturing both spatial and temporal traffic patterns. The proposed system achieves 99.5% accuracy and a 97.7% F1-score, demonstrating a significant improvement over baseline ML and DL approaches. In addition, a lightweight and scalable mitigation module embedded in the SDN controller dynamically drops or reroutes malicious flows, enabling real-time, low-latency responsiveness. Experimental results across diverse topologies confirm the framework's scalability and applicability in real-world SDN environments.

Index Terms—CNN-LSTM, deep learning, DDoS attack, machine learning, network security, SDN security, SDN vulnerabilities.

I. INTRODUCTION

IN RECENT years, cybersecurity has become a critical requirement for ensuring the safety and stability of digital systems in all sectors of the economy [1]. The rapid expansion of technologies such as cloud computing [2], mobile devices [3], the Internet of Things (IoT) [4], and digital communication platforms [5] has significantly increased the number of entry points for potential cyberattacks on industrial control systems. Consequently, protecting systems and networks from unauthorized access [6], data breaches [7], and service interruptions [8] has become increasingly important. In addition, cyberattacks are growing in sophistication, often employing advanced techniques that can bypass traditional defense mechanisms [9]. These attacks can cause serious financial damage [10], disrupt services [11], and expose sensitive information of the users [12]. Furthermore, their impact is not limited to individuals or organizations; governments and national infrastructures are increasingly being targeted, with consequences that can be both widespread and long-lasting [13].

Among the various forms of cyber threats, Distributed Denial of Service (DDoS) attacks remain one of the most frequent and disruptive [14], [15], [16]. These attacks aim to overwhelm network infrastructure or application servers by generating excessive traffic, ultimately rendering services inaccessible to legitimate users [17], [18]. In many cases, DDoS traffic is orchestrated through large-scale botnets comprising compromised devices distributed across the globe, making it exceedingly difficult to identify or block the true source of the attack [19]. The consequences can be severe, ranging from financial loss and reputational damage to prolonged service outages and loss of life [10]. In addition, no organization is immune; both small businesses and large enterprises are increasingly targeted because of the availability of attack-as-a-service tools and low execution costs [20]. Although various detection techniques have been proposed in recent years, traditional static and signature-based approaches have proven inadequate for addressing the dynamic and evolving nature of modern DDoS attacks [21]. Consequently, there is an urgent need for intelligent and adaptive mitigation strategies that can operate effectively in instantaneous,

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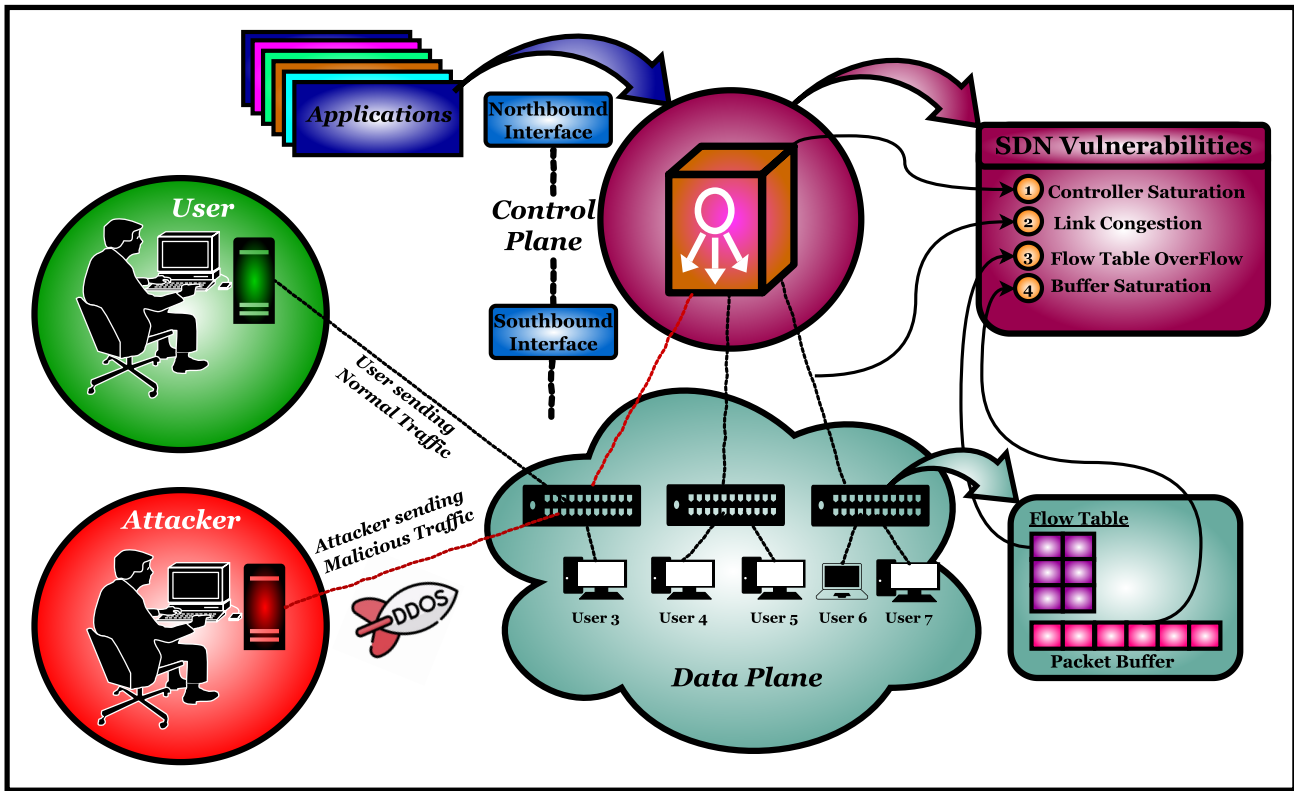


Fig. 1. Typical SDN architecture with DDoS vulnerabilities, showing normal and malicious traffic flows and attack points such as controller saturation, flow table overflow, buffer saturation, and link congestion.

high-throughput network environments [22]. Deep learning techniques are particularly well-suited for this challenge, as they can capture both fine-grained traffic features and evolving sequential patterns that are often missed by static or signature-based systems.

Software-Defined Networking (SDN) has emerged as a transformative paradigm for improving the flexibility [14], scalability [21], and security [22] of modern network infrastructures. By decoupling the control and data planes, SDN enables centralized and programmable network management, allowing administrators to monitor and configure traffic flows dynamically in response to emerging threats [23]. This centralized architecture enhances overall efficiency and supports real-time adaptation, making SDN particularly well-suited for complex and rapidly evolving threat landscapes [24]. However, despite these advantages, SDN introduces new security challenges, with DDoS attacks posing a significant threat to all three layers of the architecture: application, control plane, and data plane [25]. As illustrated in Figure 1, common vulnerabilities include buffer saturation, flow table overflow, link congestion and controller saturation. For a more comprehensive exploration of these issues, our previous work [26] presents an in-depth analysis of SDN vulnerabilities and introduces a novel taxonomy that categorizes DDoS attacks into seven distinct classes based on their targets across the SDN stack. These unique vulnerabilities highlight the need for advanced detection methods that not only analyze static flow characteristics but also adapt to evolving attack dynamics. Hybrid models such as CNN-LSTM provide this advantage

by combining spatial traffic feature extraction with sequential temporal learning, making them especially relevant for SDN environments where attacks often unfold as dynamic flow sequences.

Branches of artificial intelligence, such as machine learning (ML) and deep learning (DL), have demonstrated strong potential for detecting DDoS attacks by learning network traffic patterns [27], [28], [29]. These methods are particularly effective in capturing subtle deviations in flow activities, which are often difficult to identify using conventional approaches [30], [31]. The effectiveness of DL techniques is closely tied to the quality of feature representation and the availability of large and diverse datasets for training purposes [32], [33]. Building on this synergy, we propose a hybrid deep learning model that integrates Convolutional Neural Networks (CNNs) for spatial feature extraction with Long Short-Term Memory (LSTM) networks for temporal sequence learning. This architecture is designed to capture both static flow characteristics and dynamic behavioral patterns, thereby enabling more accurate and robust detection of DDoS attacks.

To the best of our knowledge, this study is the first to present a comprehensive DDoS detection and mitigation framework for SDN environments. It integrates a custom-built dataset, advanced feature selection techniques, hybrid deep learning models, and a real-time script-based mitigation strategy. A novel dataset was developed using Mininet and the Ryu controller to capture SYN and DoS traffic in a realistic SDN environment. It contains 74,595 labeled data points. Unlike prior studies that rely on synthetic or limited

datasets, this dataset closely mirrors real-world conditions and is specifically structured for ML applications. In addition, a multistage feature selection process, including correlation analysis, univariate testing, and chi-squared filtering, was employed to identify 16 highly discriminative features. Several classification algorithms were evaluated; notably, the proposed CNN-LSTM hybrid model achieved state-of-the-art results, with 99.5% accuracy and 97.7% F1-score, surpassing existing approaches in terms of precision and robustness. Furthermore, a lightweight, script-based mitigation mechanism was implemented directly in the SDN controller. This enables real-time classification and dynamic handling of malicious traffic, allowing the system to drop or redirect harmful packets. This approach not only enhances the response speed but also reduces the reliance on external mitigation systems. Moreover, the framework was extensively validated across diverse network topologies, confirming its effectiveness, scalability, and adaptability to modern SDN infrastructures.

Roadmap: The remainder of this paper is structured as follows. Section II provides a review of existing ML/DL-based techniques for DDoS detection in SDN environments. Section III details the proposed methodology, including dataset generation and model architecture. Section IV presents the experimental setup, results, and analysis. Finally, Section V concludes the paper and outlines directions for future research.

II. RELATED WORK

Traditional network security systems often fall short in managing the increasing scale and complexity of DDoS attacks, mainly because of their limited flexibility and scalability [33], [34]. Consequently, SDN, particularly with OpenFlow support, has gained attention for its ability to provide dynamic and centralized control over network traffic [35]. This architecture enables the deployment of customized, responsive security measures tailored to specific network requirements [36], [37]. However, SDN's centralized nature of SDNs also introduces new security risks, as the controller becomes a critical point of failure and an attractive target for attackers [38]. To enhance security in SDN environments, researchers have increasingly applied ML and DL techniques. For example, Singh et al. [39] proposed a hybrid ML-based approach for detecting and mitigating DDoS attacks in SDN environments. The framework integrates Snort IDS and a hybrid Support Vector Machine Random Forest (SVM-RF) classifier for detection and applies two mitigation strategies: dropping and redirecting malicious traffic. The SVM-RF model achieved a detection accuracy of 99.1%, outperforming other standalone classifiers. Features were selected and reduced using PCA and t-SNE before training the ML models. Despite strong results, this approach remains focused on traditional ML models without exploring DL or hybrid DL architectures. The generated dataset is not publicly available (empty link provided), which makes it difficult for other researchers to verify. Moreover, although redirection is innovative, it lacks integration with adaptive feedback or intelligent path optimization.

Alashhab et al. [40] presented a dynamic framework for detecting zero-day DDoS attacks in SDN using an ensemble online machine learning (EOML) method. The

proposed approach, which integrates four OML algorithms, “BernoulliNB,” “Passive-Aggressive,” “SGD,” and “MLP,” achieved a detection accuracy of 99.26% with high precision (99.10%) and recall (99.62%). The system was validated using a self-generated SDN dataset (not publicly available), which demonstrated consistently high performance. Additionally, the OML-based Intrusion Prevention System (IPS) reduced the CPU usage and network latency while maintaining a low false alarm rate (FAR) of 0.025. Although this study presents a novel ensemble detection system, it does not explore DL architecture techniques (e.g., CNN-LSTM), which may better capture spatiotemporal traffic dynamics. Eldhai et al. [41] investigated stream-based traffic classification in SDN using optimized feature selection and adaptive learning systems. The study introduces Boruta, a robust feature selection method, and integrates it with three stream-learning classifiers: “Hoeffding Adaptive Tree (HAT)”, “Adaptive Random Forest (ARF)”, and “KNN-ADWIN”. The Boruta-based feature selection achieved 95% accuracy, outperforming traditional genetic algorithms. When combined with stream-learning models, the system maintained up to 88% F1-score (ARF), showing superior performance in dynamic environments. However, the authors focused solely on traffic classification (detection) and did not implement or propose mitigation mechanisms. Alqahtani et al. [42] proposed a modified scanning tool, “APT-SDNmap”, that mimics the “slow and low” behavior of APTs, aimed at reconstructing flow rules in OpenFlow switches. Among the five supervised ML models tested, XGBoost, RF, KNN, DT, and SVM were used. XGBoost consistently achieved the best performance, attaining 97.82% accuracy, precision, recall, and F1-score using just the top five features. Although this study makes a significant contribution by focusing on stealthy reconnaissance detection, it does not address traffic mitigation, leaving a critical gap in the protection process of the vehicle. The proposed method does not explore multi-class attack detection or hybrid techniques. Manisankar et al. [43] introduced a hybrid ML framework for both detecting and mitigating DDoS attacks in SDN environments. The approach combines RF for the initial detection of abnormal traffic with “Gradient Boosting (GB)” to classify the detected attacks into TCP, UDP, or ICMP. The method achieved an overall detection accuracy of 97%, with a precision of 96.96%, recall of 98%, and F1-score of 97.48%. The system supports automated traffic classification and response, making it viable for integration into SDN deployments. Although the technique incorporates both detection and mitigation, it relies on traditional ML frameworks and does not explore deep or hybrid deep learning architectures (e.g., CNN-LSTM), which may yield better performance. Tonkal et al. [44] combined traditional supervised ML algorithms with Neighbourhood Component Analysis (NCA) for feature selection. The aim was to build an efficient and scalable classification pipeline that reduces the computational overhead while maintaining high accuracy. The use of NCA enabled the selection of 14 most influential features from the original 23 in the public SDN DDoS dataset. The classifiers tested included k-NN, DT, Artificial Neural Network (ANN), and SVM. The DT approach achieved 100% accuracy, whereas ANN, kNN,

and SVM achieved 99.78%, 99.15%, and 98.59% accuracy, respectively. Although the study achieved high classification accuracy with traditional ML models, it did not discuss advanced DL techniques. In addition, they did not implement mitigation mechanisms, which are important for sophisticated attacks in the SDN environment. Liu et al. [45] discussed a “*feature-engineering-based intrusion detection system for SDN*” that utilizes an improved “*binary grey wolf optimization (BGWO)*” algorithm for feature selection. The selected feature subset was then trained using five supervised learning classifiers (RF, SVM, k-NN, DT, and XGBoost), and the best-performing model was deployed in the SDN controller for DDoS detection and mitigation. The RF technique, when trained on 26 features selected from the original 79 using the improved BGWO, achieved 99.13%, 98.43%, 99.92%, and an F1-score of 99.13%. The article used an outdated synthetic public dataset, CSE-CIC-IDS2018. Although the our approach achieves high accuracy using traditional ML methods, it lacks exploration of more expressive learning paradigms, such as deep or hybrid deep learning architectures, which are increasingly vital for modeling complex network behaviors. Sangodoyin et al. [46] utilized classic ML algorithms to detect and classify DDoS flooding attacks in SDN environments. The authors emphasize a cost-effective approach using a small but balanced dataset derived from SDN scenarios. The models tested included CART, k-NN, QDA, and GNB. CART delivered the best results, with a 98.7% prediction accuracy, and outperformed the other models in terms of training speed and prediction stability. The dataset was carefully crafted from a controlled SDN environment using tools such as LOIC and iperf, yielding 3,600 data points across four traffic categories. Four features were selected manually and mathematically engineered, not via automated feature selection algorithms such as Chi2 or PCA. Overall, this study presents a solid offline detection system with high accuracy; however, it lacks integration with SDN controllers, and no mitigation mechanism is implemented. Gadallah et al. [47] presented DL-based framework for detecting and mitigating DDoS attacks in SDN control and data planes. The proposed model employs a “*hybrid autoencoder Bidirectional Gated Recurrent Unit (BGRU)*” architecture to analyze traffic flow patterns and detect anomalies in real time. The AE-BGRU method achieved 99.89% and 99.91% accuracies for the control and data plane detections, respectively, with a false positive rate (FPR) as low as 0.09% and an F1-score of 99.89%. However, the authors used a generated dataset that was not available online. Although effective, the trust mechanism is rule-based and non-adaptive, lacking reinforcement or dynamic learning to evolve with network activities. Zaidoun and Lachiri [48] explored deep learning-based multi-classification of DDoS attacks in SDN using both simple and hybrid DL models. The objective is to improve detection granularity beyond the common binary classification approach by accurately identifying five specific DDoS attack types and normal traffic. This study evaluated multiple DL models under a unified experimental setup using an optimized version of the CIC-DDoS2019 dataset. Among all the algorithms, the DNN-GRU hybrid achieved the highest accuracy of 97.37%, whereas LSTM led among the simple

models at 97.15%. The F1-scores across the hybrid models ranged from 90.5% to 94.7%. Although the framework achieves excellent accuracy in multi-class DDoS detection, it lacks a mitigation component, limiting its defensive capability in SDN. Said et al. [49] combined a “*CNN-BiLSTM*” architecture with a hybrid feature selection process. It addresses the limitations of redundancy, feature noise, and imbalanced datasets by integrating optimized feature engineering with spatiotemporal DL modeling. The proposed CNN-BiLSTM model achieved 97.77% accuracy in binary classification and 97.12% in multi-class classification using the InSDN dataset. Remarkably, it achieved 100% recall and F1-score for key attack classes, including DDoS, DoS, U2R, and Botnet. The mitigation logic is controller-based and reactive but lacks the dynamic flow rule adaptation seen in production-grade mitigation systems. Haider et al. [50] proposed a CNN ensemble framework for the efficient detection of DDoS attacks in SDN environments. The system was designed to operate at the SDN control plane using flow-based input features extracted from a benchmark dataset. The ensemble CNN model achieved a detection accuracy of 99.45%, outperforming other deep learning architectures, such as RNN, LSTM, CNN-LSTM, and hybrid RL. It also demonstrated high precision, recall, and F1-score, with significantly lower false-positive rates. The study used four key flow features selected via a Random Forest Regressor from a total of 80 features in the CICIDS2017 dataset. Despite achieving excellent classification metrics, the framework is strictly detection-oriented and does not extend to proactive threat response or mitigation logic in the SDN environment. Najar and Naik [51] developed a robust CNN-based detection and mitigation framework for SDN environments by combining “*Balanced Random Sampling (BRS)*” with deep feature selection using Information Gain. The CNN achieved 99.99% accuracy in binary classification and 98.64% in multi-class classification, with an AUC of 0.9998, indicating a near-perfect performance. The algorithm was rigorously tested under three operational scenarios: attack-free, attack without mitigation, and attack with mitigation, and demonstrated a marked drop in CPU and packet overhead when the mitigation module was active. Although efficient, the CNN method could benefit from hybrid deep learning or transformer-based modules to better handle evolving attack signature areas directly targeted in the proposed hybrid framework. Zhang and Wang [52] presented a hybrid DDoS detection and mitigation framework for SDN, called “*HESS*,” which combines a lightweight entropy-based anomaly flagging system with a deep learning classifier (SSAE-SVM) to enable efficient two-stage detection. The standalone SSAE-SVM achieved an accuracy of 98.63% and a false alarm rate (FAR) of 2.13%, whereas the combined system (entropy + SSAE-SVM) achieved an accuracy of 98.76% and further reduced the FAR to 1.8% with significantly lower CPU utilization. HESS remains rule-based in its mitigation logic, lacking adaptiveness or dynamic prioritization of flows based on evolving traffic patterns. Nguyen et al. [53] proposed a hybrid detection architecture that combines a BI-LSTM deep learning model with a “*Gaussian Mixture Model (GMM)*” for open-set recognition and uses incremental learning to adapt the model as new threats are

identified. The initial BI-LSTM model trained on DoS traffic achieved an accuracy of 97.23% and a recall of 0.9979 for known attacks. However, it performed poorly on unseen attack types, such as those in the CICDDoS2019 dataset. Although the framework effectively detects both known and unknown DDoS attacks, no mitigation mechanism is incorporated into it. In addition, the detection logic is decoupled from the SDN operation, and the performance of the GMM in high-speed, SDN settings has not been benchmarked. Elsayed et al. [54] presented “*DDoSNet*,” a deep learning-based intrusion detection system specifically designed for SDN environments. The presented system combines an RNN-based autoencoder (AE) with a softmax classifier to perform binary classification of network traffic as either benign or malicious. DDoSNet achieved 99% accuracy, precision, recall, and F1-score, demonstrating strong detection capabilities. The area under ROC curve (AUC) reached 98.8, indicating high separability between benign and malicious traffic. There is no flow rule-level integration with the SDN controller to enforce traffic control. Vajrobol et al. [55] proposed method combines the temporal learning capabilities of LSTM with the gradient-boosting precision of XGBoost, further strengthened by adversarial training to resist manipulation and evade detection attempts. The model was trained and tested on real-world Mirai variants using the CICIoT2023 dataset. The hybrid model of LSTM + XGBoost achieved an overall accuracy of 97.7% with strong precision, recall, and F1-scores across four Mirai attack types (GREETH, GREIP, UDP-Plain, and benign). However, the framework remains detection-centric and does not integrate any autonomous mitigation mechanisms. The system effectively identifies malicious patterns but does not extend to active response actions, such as traffic shaping, flow rule enforcement, or packet redirection, within an SDN controller. Mhamdi and Isa [56] introduced deep autoencoder-based feature learning with a Random Forest classifier. It is deployed inside the SDN controller as part of a three-layer adaptive architecture that integrates entropy-based anomaly detection and passive service monitoring for layered decision-making. The DAERF model achieved 98% accuracy using only eight selected OpenFlow-native features. When integrated into the full DAERF_IDS three-layer framework, the system achieved an accuracy of 98.16% accuracy, 98.92% F1-score, and a false-positive rate of 1.85%. However, mitigation is layered and adaptive, and the response logic is threshold-based rather than predictive or reinforcement-driven. Pradeesh et al. [57] integrated modular detection strategies, including signature-based, anomaly based, and behavior-based models, combined with “*adaptive ensemble learning (AEL)*” and dynamic weighting using “*One-vs-Rest (OvR)*” classifiers. The proposed “*Adaptive Ensemble of Modular Classifiers (AEMC)*” framework achieved a classification accuracy of 96.7%, with precision, recall, and F1-score all at 0.967. Additional metrics further confirmed its robustness: an AUC-ROC of 0.97, FPR of 0.04, and FNR of 0.03. However, the technique mitigation logic is limited to threshold-based decision-making without demonstrating the actual enforcement of flow rules or traffic rerouting via the SDN controller.

Recent works have also investigated CNN-LSTM hybrids for DDoS detection in SDN and related domains. For instance, Ullah et al. [58] proposed an explainable CNN-LSTM model for IoT-enabled Healthcare 5.0, achieving very high accuracy on CICDDoS2019 but relying primarily on public datasets without addressing controller-level deployment challenges. Jimoh et al. [59] explored CNN-LSTM for SDN DDoS mitigation, reporting strong results on ISCXIDS2012, though their work was limited to an older dataset and lacked validation under realistic SDN emulation or custom traffic. Afroj et al. [60] developed ensemble and hybrid CNN-LSTM models, showing that stacking ensembles outperformed standalone hybrids, but their study remained offline and did not integrate mitigation within the controller. Similarly, a two-level CNN-LSTM detection framework was introduced in [61], demonstrating promising accuracy, but focusing primarily on architectural layering without considering feature redundancy or in-controller resource constraints. While these studies confirm the utility of CNN-LSTM models, they predominantly emphasize accuracy on benchmark datasets and give limited attention to (i) dataset realism/diversity tailored to SDN, (ii) principled feature reduction to avoid redundancy and reduce overhead, and (iii) integrated, controller-level mitigation with measured resource impact. In contrast, our framework couples a *custom SDN-specific dataset* with a *multi-stage feature selection pipeline* and *in-controller mitigation*, expressly targeting both detection quality and deployment feasibility.

Recent studies have advanced DDoS detection in network environments. Yang et al. [62] detected MAWD attacks in high-speed networks with over 99% recall, while authors in [63] proposed a ML model for detecting DDoS attacks with system variables, achieving 58%-95% accuracy. A tensor decomposition framework for DDoS detection in big data environments achieved 99.19% accuracy [64], and P4-Secure introduced an in-band DDoS detection system in the data plane, minimizing false positives to 0.5% [65]. Our study differs by using a hybrid CNN-LSTM model for DDoS detection in SDN environments with integrated real-time mitigation in the SDN controller, offering enhanced scalability and responsiveness.

Recent contributions have also explored DDoS detection in SDN-enabled and IoT-driven environments. For instance, [66] proposed a dynamic multiphase framework for SDN-based fog-enabled IoT networks that combines entropy-based anomaly detection, statistical thresholding, and machine learning classifiers such as Random Forest. Their approach achieved high accuracy (98.71%) on the BoT-IoT dataset and demonstrated reduced response time through fog-layer deployment. Similarly, [67] utilized statistical filtering combined with ML classifiers on the CICDDoS2019 dataset to predict variants of DDoS attacks, achieving effective traffic reduction and attack-type identification. While both works confirm the utility of statistical and ML-based methods, they remain reliant on entropy thresholds or shallow classifiers and focus primarily on offline datasets. In contrast, our study introduces a hybrid CNN-LSTM model with a custom SDN-specific dataset, employs a multi-stage feature selection process, and integrates real-time mitigation directly inside the controller,

TABLE I

SUMMARY AND COMPARATIVE VIEW OF RELATED WORKS (M: MITIGATION, D: DETECTION). THE TABLE EMPHASIZES DIFFERENCES IN TECHNIQUE, DATASET CHOICE, PERFORMANCE REPORTING, AND WHETHER MITIGATION AND DEPLOYMENT CONSIDERATIONS WERE ADDRESSED

Reference	Techniques	Features	Performance	Dataset	M	D
<i>Singh et al. [39]</i>	SVM+RF	12 features	Accuracy 99.1%	Self-Generated	✓	✓
<i>Alashhab et al. [40]</i>	EOML	14 features	Accuracy 99.26%	Self-Generated	✓	✓
<i>Wldhai et al. [41]</i>	ARF+KNN	10 features	Accuracy 95%	Self-Generated	✗	✓
<i>Alqahtani et al. [42]</i>	XGBoost, RF, KNN, DT, SVM	5 features	Accuracy (XGBoost) 97.82%	Self-Generated	✗	✓
<i>Manisankar et al. [43]</i>	RF+GB	7 features	Accuracy 97%	Self-Generated	✓	✓
<i>Tonkal et al. [44]</i>	ML+NCA	8 features	Accuracy (DT) 99.17%	Self-Generated	✗	✓
<i>Liu et al. [45]</i>	BGWO + ML	26 features	Accuracy (RF) 99.13%	CSE-CIC-IDS2018	✓	✓
<i>Sangodoyin et al. [46]</i>	ML-CFA	8 features	Accuracy (CART) 98.9%	Self-Generated	✗	✓
<i>Gadallah et al. [47]</i>	AE+BGRU	8 features	FAR 90%	Self-Generated	✓	✓
<i>Zaidoun et al. [48]</i>	DNN+GRU	70 features	Accuracy 97.15%	CIC-DDoS2019	✗	✓
<i>Said et al. [49]</i>	CNN + BiLSTM	10 features	Accuracy 97.77%	Self-Generated	✓	✓
<i>Haider et al. [50]</i>	CNN Ensemble+RF	4 features	Accuracy 99.45%	CICIDS2017	✗	✓
<i>Najar & Naik [51]</i>	CNN+BRS	58 features	AUC: 0.9998	CIC-DDoS2019	✓	✓
<i>Zhang & Wang [52]</i>	SSAE+SVM	17 features	Accuracy 98.63%	Self-Generated	✓	✓
<i>Nguyen et al. [53]</i>	BI-LSTM + GMM	Not specified	Accuracy 97.23%	CICIDS2017	✗	✓
<i>Elsayed et al. [54]</i>	RNN+AE	77 features	Accuracy 99%	CICDDoS2019	✗	✓
<i>Vajrobol et al. [55]</i>	LSTM + XGBoost	77 features	Accuracy 97.7%	CICIoT2023	✗	✓
<i>Mhamdi & Isa [56]</i>	AE+RF	8 features	Accuracy 98.0%	Self-Generated	✓	✓
<i>Pradeesh et al. [57]</i>	AEL+ Ensemble OvR	Not specified	Accuracy 96.7%	Self-Generated	✓	✓
<i>Ullah et al. [58]</i>	CNN-LSTM + SHAP (Explainable AI)	Not specified	ACC: 99.59%, 98.12%	CICDDoS2019, IoT Healthcare Security	✗	✓
<i>Jimoh et al. [59]</i>	CNN-LSTM	Not specified	Accuracy 99.52%	ISCXIDS2012	✓	✓
<i>Afroj et al. [60]</i>	Hybrid CNN-LSTM + Ensemble (Voting/Stacking)	23 features	Accuracy 98.51%	Custom SDN dataset (104k flows)	✗	✓
<i>Li et al. [61]</i>	Two-level CNN-LSTM	Not specified	ACC: ~98%; PRC: ~97%; RCL: ~97%	Custom SDN traces	✓	✓

thereby addressing both detection accuracy and deployment feasibility in operational SDN environments.

A. Research Gap

After conducting a comprehensive literature review using a systematic search strategy, we examined research articles published between 2021 and 2025. The search was primarily focused on high-impact journals indexed in the Web of Science (WoS), including IEEE Xplore, ScienceDirect, Springer, and MDPI. From an initial pool of 35 articles closely aligned with our research objectives, we applied rigorous screening criteria and selected 25 key studies that were directly related to our research domain. These studies provide valuable insights into the current state of DDoS detection and mitigation in SDN. However, a critical evaluation revealed several research gaps that remain largely unaddressed. These gaps not only limit the practical deployment of existing solutions but also highlight the need for a more robust and scalable framework capable of providing instantaneous detection and responses. Motivated by these gaps, our research aims to develop a comprehensive, reliable, and effective detection and mitigation solution tailored for SDN environments. We have listed the identified research gaps in the existing studies as follows:

- Most existing studies (see Table I) rely on synthetic or outdated datasets that do not accurately reflect modern SDN traffic or real-world DDoS attack scenarios. In many

cases, the datasets used are not publicly accessible, making reproducibility and benchmarking extremely difficult for other researchers [39], [40], [41], [42], [43], [46], [47], [49], [52] and [56].

- Many studies use full feature sets without applying rigorous feature selection methods, which can result in overfitting, increased computational cost, and reduced model interpretability [41], [46] and [53].
- Although DL has been applied to SDN security, there is a noticeable lack of hybrid architectures, such as CNN-LSTM, that can effectively capture both spatial and temporal features in network traffic patterns [39], [40], [42], [43], [44] and [45].
- Most studies focus solely on detection, with few addressing integrated and lightweight mitigation strategies that can operate within an SDN controller in real time [41], [42], [44], [46], [48], [50], [53], [54] and [55].
- A large proportion of research validates models on simplified or static network topologies, limiting confidence in their adaptability to diverse and large-scale SDN infrastructures [40], [42], [43] and [46].
- Many frameworks depend on third-party tools to block or redirect malicious traffic, which introduces latency and additional dependencies, undermining the responsiveness of the system [39] and [40].
- Current research tends to address either detection or mitigation in isolation, with minimal effort toward building

TABLE II
FEATURES OF THE DATASET

Ser No.	Feature Name	Description
1	Flow Identifiers	flow_index, source_ipv4, dest_ipv4, source_port, destination_port
2	Protocol Info	ethernet_type, ip_protocol
3	Volume Stats	total_packets, flow_bytes_per_sec, flow_packets_per_sec, avg_packet_size
4	Header Info	fwd_header_length
5	Timing (Bytes)	min_bytes_interval_2s, mean_bytes_interval_2s
6	Timing (Packets)	min_packets_interval_2s, mean_packets_interval_2s, std_packets_interval_2s
7	Activity Duration	flow_duration_sec
8	Active State	active_max_time, active_mean_time, active_min_time, active_std_time
9	Idle State	idle_max_time, idle_mean_time, idle_min_time, idle_std_time
10	IP Behavior Stats	total_bytes_per_dst_ip, total_bytes_per_src_ip, total_pkts_per_dst_ip, total_pkts_per_src_ip, total_pkts_per_dst_port, total_flows_per_dst_ip, total_flows_per_src_ip
11	Labels	Category, label

holistic frameworks that integrate both capabilities for real-time SDN defence [41], [42], [44], [46], [48], [50], [53], [54] and [55].

In response, our work (i) constructs and releases a custom SDN dataset, (ii) employs principled, multistage feature selection to reduce redundancy and controller overhead, and (iii) integrates detection with in-controller mitigation while reporting resource and latency observations. These choices directly target the deployment-oriented gaps surfaced by the comparative analysis.

Comparative Synthesis: Across the surveyed literature, three trends emerge. *First*, many studies achieve high accuracy on public datasets (e.g., CIC* family) but seldom assess controller-level deployability or resource overheads, limiting low-latency applicability in SDN. *Second*, mitigation is either absent or rule-based in most works, with few integrating detection and response inside the controller to minimize reaction latency. *Third*, several approaches rely on full feature sets or ad-hoc selection, which can inflate computation and obscure interpretability compared to principled, multistage filtering. Table I consolidates these differences, highlighting gaps in dataset realism/diversity, in-controller mitigation, and resource-awareness that motivate our design choices. Additionally, our custom dataset, developed specifically for SDN environments, stands as a critical differentiator. Unlike many existing studies that rely on outdated or synthetic datasets, our dataset accurately mirrors real-world SDN traffic and is freely available for future research, ensuring reproducibility and enabling broader benchmarking of DDoS detection techniques.

III. METHODOLOGY AND DATASET DEVELOPMENT

Figure 2 illustrates the workflow designed for DDoS-attack detection within an SDN. The process began with the creation and structuring of a custom CSV-formatted dataset. This is followed by a data inspection and exploration (DIE) phase that identifies the underlying patterns, statistical properties, and any label imbalance using correlation metrics and visual analytics. Based on these insights, feature selection was conducted using techniques such as correlation filtering, univariate testing, and

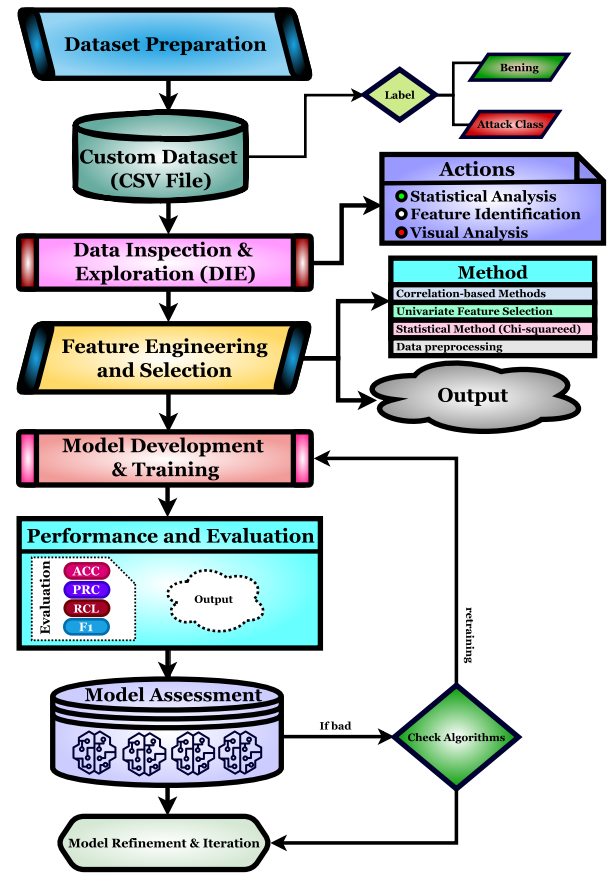


Fig. 2. Design framework of the proposed model.

chi-squared analysis to isolate the most predictive attributes. These features were used to train several algorithms, including LR, NB, LSTM, and hybrid CNN-LSTM. The trained frameworks were then evaluated using standard metrics, such as accuracy, precision, recall, and F1-score. The results guide a refinement loop, wherein algorithm selection, hyperparameter tuning, and feature configurations are iteratively adjusted to enhance the detection performance.

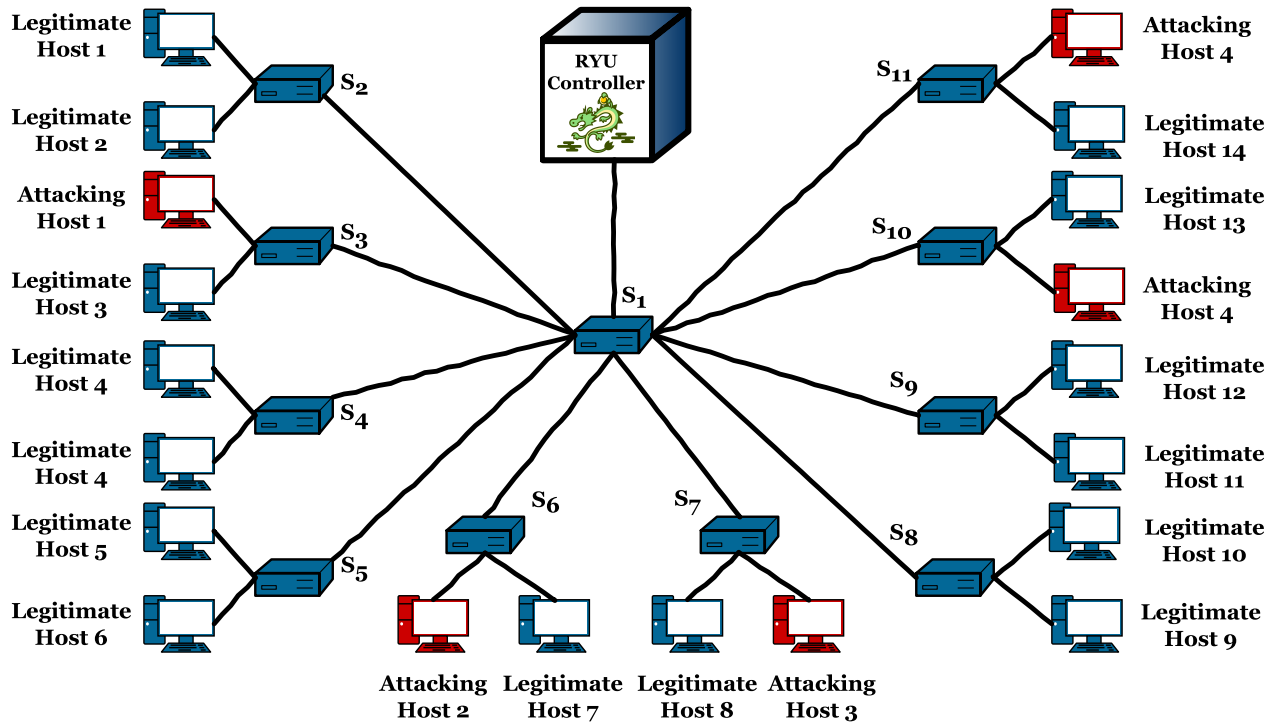


Fig. 3. Experimental topology used for DDoS attack simulation in the SDN testbed.

A. Dataset Generation

A controlled experimental environment was established to develop a robust dataset for DDoS detection in SDNs. Specifically, Mininet served as the emulation platform, and the RYU controller handled the SDN logic. The network topology illustrated in Figure 3 comprises multiple OpenFlow-enabled switches connecting various end hosts, including both benign and malicious nodes. This setup was designed to reflect real-world distributed systems with a central controller that manages all flow decisions. To simulate attack scenarios, malicious hosts launch different types of DDoS attacks, such as SYN floods, ICMP floods, and UDP-based volumetric attacks, targeting one or more designated victim servers. These attacks were generated using tools such as hping3, Scapy, and custom traffic scripts. Legitimate hosts maintain regular communication patterns. This ensured that the dataset represented a realistic blend of normal and malicious traffic. Traffic flow was captured at the switch level using OpenFlow counters supplemented by additional monitoring scripts. The collected data were exported in a structured format *.xlsx* files, with separate entries for attack and benign traffic. Each record includes key attributes, such as the packet count, byte rate, flow duration, inter-arrival time, and per-IP activity metric. Forty initial features were extracted from raw data. A correlation-based feature selection technique was applied to enhance model efficiency and reduce redundancy. Consequently, 35 relevant features were retained, representing flow characteristics, volume trends, and timing scenarios. A binary label (0 for normal and 1 for an attack) was added to each entry to support supervised learning. The final dataset included 74,595 samples that were evenly balanced across

the attack traffic classes. These features include span flow identifiers, protocol types, traffic volumes, timing intervals, activity durations, and host behaviors. The complete list of retained features is grouped and summarized in Table II. This comprehensive and well-structured dataset was subsequently used to train and evaluate a range of ML and DL techniques for effective DDoS detection in an SDN environment.

It is important to note that while our dataset incorporates multiple network- and transport-layer attack types (e.g., SYN floods, UDP floods, ICMP floods, scanning, and password-guessing), the present study focused on volumetric DDoS attacks. Application-layer and low-rate stealthy attacks, such as HTTP GET/POST floods or slowloris, were not yet included in this version of the dataset. These attack categories pose unique detection challenges and will be addressed in future dataset extensions and evaluation studies.

B. Preprocessing

Data preprocessing is crucial in AI because it directly affects model performance [68]. This step transforms raw data into a structured format suitable for training, ensuring that the framework learns from clean and relevant information. Inadequate preprocessing, such as ignoring noise, inconsistencies, or irrelevant data, can lead to inaccurate and overfitted systems. Effective preprocessing enhances both the predictive accuracy and the algorithm's ability to generalize. One significant challenge in preprocessing is addressing missing or zero values, which may arise from incomplete data or measurement errors [1]. These values can bias the method; therefore, they must be handled with care. In this study, columns with missing or zero values that did not provide meaningful information were removed to retain only

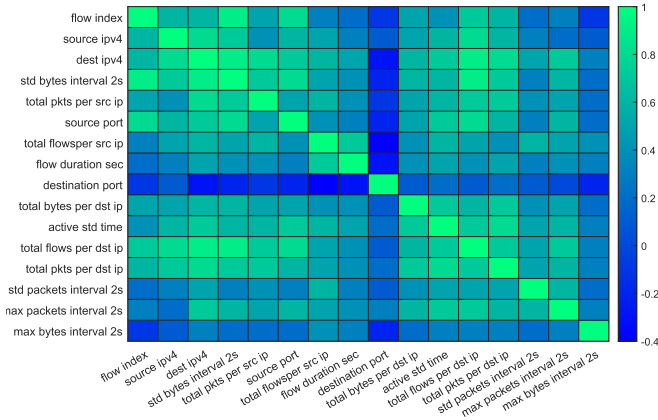


Fig. 4. Correlation Among Features.

the relevant features, minimize the risk of overfitting, and improve generalization. Feature engineering involves selecting the most informative attributes that best represent the underlying patterns in the data. By focusing on relevant features, such as “*source_ipv4*” or “*total_pkts_per_src_ip*,” we ensured that the classifier was trained on the most valuable information, minimizing redundancy and improving accuracy. This step is crucial for preventing the overfitting of unseen data. To refine feature selection, statistical methods such as correlation and chi-squared tests were employed to assess the relevance of each feature. Features that exhibit strong correlations (either positive or negative) are deemed more significant because they contribute meaningfully to the predictive power of the model.

Three feature selection techniques were applied in this study: SelectBest, which automatically identifies the top features based on predefined criteria; Univariate Selection, which targets features that are most strongly related to the target variable; and Correlation-Based Selection, which evaluates the relationships between features to eliminate redundancy. By applying these methods, we retained the most informative features to improve system accuracy. A heat map, as shown in Figure 4, was used to visualize the feature correlations. The color coding represents the strength of the relationships: darker shades of blue indicate strong positive correlations, whereas darker shades of green and light blue represent weak or no correlations. Lighter blue and green shades signify weaker negative correlations, with the most intense blue hue representing a stronger negative correlation. This visualization aids in identifying the key feature pairs for inclusion or exclusion in the model. Finally, five algorithms were applied to evaluate the 35 dataset features. This multi-algorithm approach ensured that only the most relevant features were retained, optimizing the dataset for model training and improving the overall accuracy.

C. DDoS Attacks

In this study, we used a novel dataset encompassing various attack types, including DoS, DDoS, scanning, and password-guessing attacks, which represent a broad spectrum of threats to network security. However, this study specifically focuses on DDoS attacks, which continue to be among the most

TABLE III
SELECTED FEATURES

Ser No.	Selected Features
1	flow_index
2	source_ipv4
3	dest_ipv4
4	std_bytes_interval_2s
5	total_pkts_per_src_ip
6	source_port
7	total_flows_per_src_ip
8	flow_duration_sec
9	destination_port
10	total_bytes_per_dst_ip
11	active_std_time
12	total_flows_per_dst_ip
13	total_pkts_per_dst_ip
14	std_packets_interval_2s
15	max_packets_interval_2s
16	max_bytes_interval_2s

widespread and disruptive forms of network intrusion [52]. Unlike traditional DoS attacks, DDoS attacks leverage multiple systems or botnets to flood a target with excessive traffic, making detection and mitigation significantly more challenging [56]. Although our dataset includes multiple attack vectors, it is particularly suitable for applying ML/DL methods to identify and classify DDoS traffic patterns. The primary aim of our analysis was to enhance the detection and mitigation of DDoS threats in SDN environments.

D. Feature Selection

Effective training and evaluation of ML/DL techniques rely heavily on the availability of high-quality and well-structured datasets to ensure reliable and valid performance assessment [49], [69]. In this study, highly correlated features were identified and removed using a threshold-based correlation matrix, thereby mitigating multicollinearity and preserving only the independent variables. Subsequently, a feature importance analysis was conducted to evaluate the contribution of each feature to classification performance. Based on this dual-stage refinement, a concise set of 16 features was finalized. This curated set encapsulates the essential characteristics of network traffic across multiple dimensions, including volume metrics, timing, flow-level attributes, and host-specific communication patterns. Importantly, features such as “*flow_duration_sec*,” “*avg_packet_size*,” and “*flow_bytes_per_sec*” were particularly influential in distinguishing between benign and malicious flows. Table III provides an overview of the features selected based on their relevance and contribution to detection tasks.

Although DL models are capable of automatic feature learning, applying explicit feature selection is particularly beneficial in SDN-based DDoS detection. First, flow-level datasets often include highly correlated and noisy attributes (e.g., multiple counters per IP), which can mislead the learning

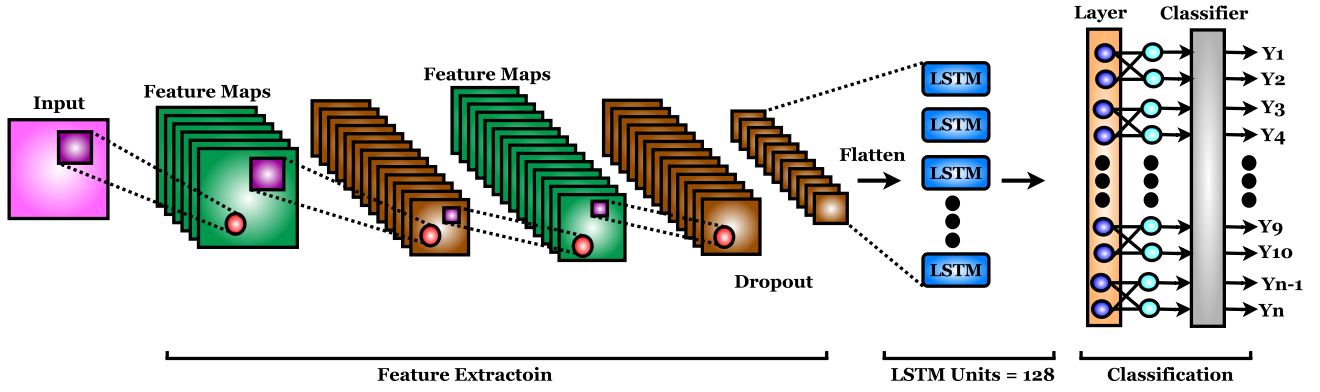


Fig. 5. Architecture of hybrid Model.

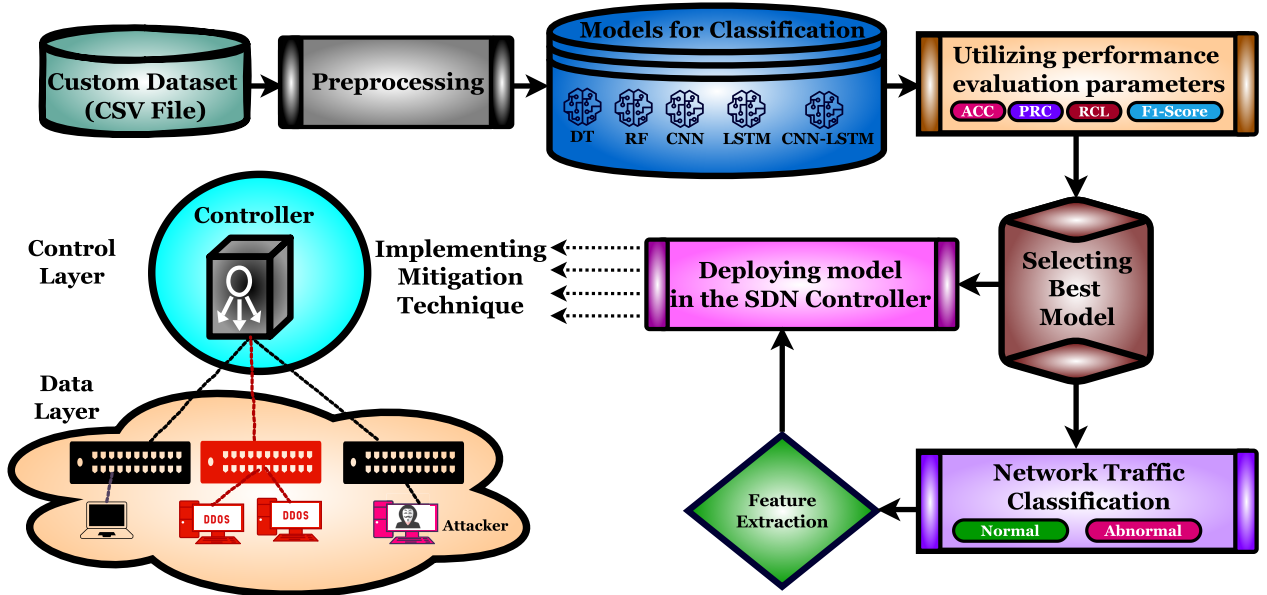


Fig. 6. Proposed ML/DL-based intrusion detection framework.

process; eliminating such redundancy reduces the risk of overfitting, especially given that our dataset size (74k samples) is modest by deep learning standards. Second, minimizing the feature set lowers the computational cost of both training and instantaneous inference, which is critical when models are deployed inside resource-constrained SDN controllers that must process flows with minimal latency. Third, feature selection improves interpretability by highlighting the specific traffic attributes (e.g., duration, packet size, flow rates) that most strongly contribute to attack detection, offering actionable insights for network operators. To achieve these benefits, we adopted a complementary three-step strategy: correlation analysis removed multicollinearity, univariate filtering assessed discriminative power, and chi-square testing verified statistical association with the attack label. This multi-stage pipeline ensured that the final 16 features preserved maximum predictive value while supporting scalability, robustness, and deployment feasibility.

For correlation filtering, a threshold of 0.8 was selected based on prior SDN traffic studies, as values above this level indicate strong multicollinearity that may obscure model learning while lower thresholds risk discarding informative features. Similarly, chi-square testing was applied to evaluate the statistical dependence of categorical features with the class label, ensuring that only features with meaningful discriminative power were retained. These choices balance the trade-off between reducing redundancy and preserving predictive capability, thereby improving transparency and reproducibility of the feature engineering process.

E. Model Design and Architecture

This study presents an advanced framework that integrates ML/DL techniques to improve the accuracy and efficiency of network IDSs, which play a critical role in identifying security threats. The process began with a comprehensive

analysis of the dataset, including the inspection of structure, data types, and key statistical characteristics. To enhance pattern recognition, categorical features were transformed using OneHotEncoding. Next, the data were prepared to ensure the proper configuration of the input features and target labels for training. Several algorithms have been employed, including DT, RF, CNN, LSTM networks, and a combined CNN-LSTM model. These approaches were evaluated using confusion matrices and classification metrics, such as precision, recall, accuracy, and F1-score. Notably, the hybrid CNN-LSTM architecture combines temporal and spatial feature extraction, resulting in improved predictive performance. Rule-based methods, such as RF and DT, offer complementary strengths that can further enhance classification accuracy. Extensive testing confirmed the robustness and effectiveness of the proposed approach in detecting abnormal traffic patterns and potential intrusions. As illustrated in Figure 6, the hybrid method begins with preprocessing and feature extraction from a custom dataset. The processed data were then used to train and evaluate multiple frameworks. The best-performing approach was deployed in the SDN controller for traffic classification and attack mitigation. This layered architecture includes a control and data plane, ensuring detection and response capabilities within the network environment. Furthermore, we chose the following techniques:

1) *Decision Trees (DT)*: DT emulate human decision-making processes, which contributes to their intuitive design and ease of interpretation [70]. These algorithms are structured as trees, where each internal node evaluates a specific feature and the branches represent the possible outcomes of that feature. A key advantage of DTs is their robustness to variations in feature scales, allowing them to perform effectively even when the input data are not normalized. Additionally, their framework is independent of the actual feature values, making it particularly well-suited for handling non-normalized datasets. The effectiveness of the DT approach is evidenced by its accuracy of 96.8% on the given dataset, demonstrating strong performance in this context.

2) *Random Forest (RF)*: The RF technique employs an ensemble learning strategy, utilizing multiple decision trees to perform classification tasks [71]. Each tree contributes to the final prediction, collectively enhancing the reliability of the results. In RF, the overall accuracy is determined by averaging the results from all individual classifiers, which helps to reduce the influence of weaker learners. This ensemble approach ensures that high-performing trees dominate the decision-making process, thereby improving predictive accuracy. RF is particularly effective for datasets with numerous significant features because it handles complex data structures with ease. When tested on our custom dataset, this approach achieved a notable accuracy rate of 97.1%, thus highlighting its strong performance.

3) *Convolutional Neural Networks (CNNs)*: CNNs are widely recognized for their effectiveness in image classification; however, one-dimensional convolution (Conv-1D) has also shown notable success in areas such as anomaly detection and recommender systems [72]. A central element of CNNs is the convolutional layer, which uses kernels to extract

meaningful patterns from the input data. This process involves applying kernel weights across the input vector to highlight relevant characteristics and is mathematically expressed as:

$$o_t = \tanh(y_t * h_t + s_t) \quad (1)$$

where o_t denotes the output, y_t is the input, h_t represents the kernel weights, and s_t is the bias term. Subsequently, a max-pooling layer compresses the feature representation, retaining key information while reducing dimensionality. Conv-1D is particularly effective in identifying important features for classification tasks and often requires minimal data preprocessing. When applied to our custom dataset, the Conv-1D architecture achieved a remarkable accuracy of 98.2%, highlighting its strong performance and suitability for the task.

4) *Long Short-Term Memory (LSTM)*: LSTM networks were introduced to overcome a key drawback of standard Recurrent Neural Networks (RNNs): their difficulty in retaining long-term dependencies. Traditional RNNs suffer from the vanishing gradient problem, where diminishing gradients hinder effective learning in the earlier layers, making it challenging to model sequences with long-range temporal patterns. LSTMs address this limitation through a gating mechanism comprising input, forget, and output gates that dynamically manage the information retained or discarded over time. This architecture enables the network to preserve crucial sequence data and make informed predictions over an extended time step. An LSTM unit is typically described by the following equation:

$$c(t) = f_t * c_{t-1} + i_t * c_t \quad (2)$$

where:

- $c(t)$ is the state of a **memory cell** at time t ,
- f_t is the **forget gate** output
- c_{t-1} is the previous **cell state**
- i_t is the **input gate**.

Owing to this structure, LSTMs are highly effective for modeling time-series data and sequences with such dependencies. However, in this evaluation, the model fell short of expectations, achieving an accuracy of 95%. This performance lag, compared to the other models in the study, indicates potential difficulties in capturing dataset-specific patterns or insufficient tuning for optimal feature extraction.

5) *CNN-LSTM*: The CNN-LSTM hybrid architecture incorporates a set of essential hyperparameters that significantly influence its overall effectiveness. In the first convolutional layer, 32 filters were employed to extract features from the input data with a kernel size of 3×3 , enabling the model to detect localized patterns across various spatial scales. A second convolutional layer followed, utilizing 64 filters with the same kernel size. To introduce nonlinearity and enhance the learning capacity of the model, the ReLU activation function was applied after each convolutional operation was performed. In addition, “same” padding was used to maintain the original spatial dimensions throughout the convolutional process. Each convolutional layer is followed by a 2×2 max-pooling layer that reduces the spatial dimensions and aids in feature generalization. Subsequently, a dropout layer is introduced to reduce overfitting by randomly deactivating neurons during

TABLE IV
CNN-LSTM HYPERPARAMETER CONFIGURATION

Parameter	Value
Conv1 Layer Filters	32 (kernel size 3×3, ReLU, same padding)
Conv2 Layer Filters	64 (kernel size 3×3, ReLU, same padding)
Pooling Layers	Max-pooling 2×2 after each conv layer
Dropout Rate	Applied after pooling
LSTM Hidden Units	128 (tanh activation)
Batch Size	64
Learning Rate	0.001
Optimizer	Adam
Training Epochs	50

training. The resulting feature maps were then flattened into a 1D vector before being fed into the LSTM component. Within the LSTM layer, 128 hidden units were configured to capture the temporal relationships and long-term dependencies across the sequential data. The tanh activation function was adopted to manage state transitions and regulate the memory flow within the LSTM cells. For training purposes, a batch size of 64 was used, allowing the model to update its weights after processing each group of samples. The learning rate was set to 0.001, which determined the pace of weight adjustments during optimization. Adam was selected as the optimizer, offering adaptive learning rate adjustments that contribute to more efficient convergence. The training was conducted over 50 epochs, providing ample iterations for fine-tuning and performance improvement. Figure 5 illustrates the structural composition of the CNN-LSTM model, including the layer configuration, pooling operations, dropout integration, and the final classifier. This diagram enhances the understanding of the internal workflow of the model, from the initial spatial feature extraction to sequential data interpretation and classification. Notably, the hybrid model achieved an impressive accuracy rate of 99.5%, underscoring its high effectiveness for the given task.

6) *CNN-LSTM*: The CNN-LSTM hybrid combines convolutional layers for spatial feature extraction with LSTM units for modeling sequential dependencies in flow-level traffic. Figure 5 illustrates the architecture, where convolutional and pooling layers capture localized traffic patterns, dropout reduces overfitting, and the LSTM models temporal relationships before final classification. The complete hyperparameter configuration is summarized in Table IV.

F. Mitigation

To address DDoS attacks in an SDN environment, a script-based mitigation solution was developed, as shown in Algorithm 1. This solution incorporates a proactive mechanism known as the “Drop Method,” which is designed to swiftly identify and block illegitimate traffic by targeting the source IP address. As illustrated in Figure 7, the method begins with the traffic reception and inspection by an IDS. Upon detecting potential DDoS indicators, the system conducts a detailed examination of incoming data packets. If malicious activity is confirmed, the system isolates the offending packets based on their origin and promptly rejects them, preventing them

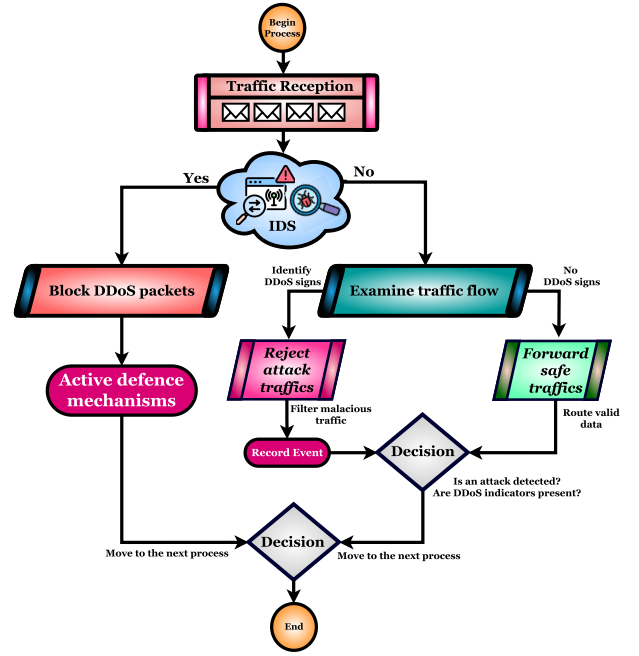


Fig. 7. Flowchart of the DDoS mitigation process in the SDN controller.

from overwhelming the network or reaching their intended destinations. This method not only strengthens security but also ensures efficient traffic management through continuous monitoring and precise packet filtering of the data packets.

Moreover, the script-based approach significantly enhances the operational efficiency within the SDN framework. Unlike traditional solutions that often rely on external applications and third-party modules, introducing latency and increasing resource consumption, this method is fully integrated into the network’s control logic. Such integration enables the streamlined processing of traffic, allowing the system to filter malicious data with minimal processing power and memory requirements. Conventional filtering techniques, which struggle with the computational overhead of analyzing large volumes of packet data, are far less efficient than the proposed method. In contrast, the proposed solution effectively mitigates this burden, enhancing both the resilience of the network to DDoS attacks and the performance of the underlying infrastructure. By balancing robust security with efficient resource use, this method ensures that both protection and operational efficiency are maintained.

While packet dropping and redirection may appear simplistic compared to more advanced approaches such as traffic shaping or adaptive QoS enforcement, these mechanisms were deliberately chosen to demonstrate the feasibility of integrating detection and mitigation directly within the SDN controller. Their lightweight design ensures minimal latency and resource overhead, which is crucial for validating operation. Furthermore, the scripts are modular and can be extended to incorporate more sophisticated strategies in future iterations of this framework.

To quantify the efficiency of the mitigation process, we monitored controller resource usage and traffic load reduction during operation. When the CNN-LSTM classifier

Algorithm 1 DDoS Traffic Mitigation With Reverification and Rollback

Input: *Network traffic flows with extracted features (flow_index, src_ip, dst_ip, ports, byte/packet rates, duration, etc.)*

Output: Classified flows (Normal or Malicious) with mitigation actions.

Initialization: Define thresholds, detection window = 2s, verification interval = 3 windows.

```

1: Step 1: Receive and preprocess traffic
2: foreach flow ido
3:   Extract flow features
4:   Store metadata for detection
5: end for
6: Step 2: Detect suspicious behavior
7: if flow i exceeds anomaly thresholds then
8:   Flag as suspicious and apply temporary rate-limiting
9:   Log flow for secondary verification
10: end if
11: Step 3: Reverification across windows
12: if suspicious flow persists for  $\geq 3$  intervals then
13:   Escalate to blocking rule (drop packets from src_ip)
14: else
15:   Rollback and restore flow if flagged traffic behaves normally
16: end if
17: Step 4: Spoofing safeguards
18: if src_ip suspected of spoofing then
19:   Cross-check consistency (flow duration, byte/packet ratios, bidirectional symmetry)
20:   Block only if anomalies are confirmed
21: end if
22: Step 5: Forward legitimate flows
23: if no DDoS indicators remain then
24:   Forward traffic normally
25: end if

```

triggered the mitigation script, the controller's CPU utilization remained below 12% and memory usage below 9% of available resources, confirming lightweight integration. Redirecting malicious traffic resulted in a measurable decrease in bandwidth consumption and packet rates on the primary links, thereby alleviating network strain while maintaining service availability for legitimate users. The end-to-end delay from detection to rule enforcement was negligible, demonstrating that the mitigation can be executed in real time without degrading controller performance.

To further minimize the risks of false positives and IP spoofing, Algorithm 1 was enhanced with a verification and rollback mechanism. Suspicious flows are first rate-limited rather than blocked, and their behavior is re-evaluated across three consecutive detection windows. Only if anomalies persist is a permanent block rule enforced. This staged approach prevents service interruptions due to transient anomalies. In addition, potential spoofed IP addresses are validated using flow-level consistency checks, such as packet-to-byte ratios and bidirectional symmetry. If legitimate behavior is detected,

the system automatically rolls back the block rule, restoring service continuity. This layered decision logic ensures that blocking is applied only when malicious behavior is confirmed, while legitimate users remain unaffected.

In our setup, "real-time" refers to the ability of the SDN controller to classify and act on flows within milliseconds of their arrival. Specifically, flows were sampled at 2-second intervals, features were extracted in 1.2 ms per flow, and CNN-LSTM inference required 3.8 ms per flow on CPU. Overall end-to-end latency, including mitigation, remained below 15 ms, while controller CPU and memory usage stayed under 12% and 9%, respectively.

G. Experimental Setup and ML Techniques

The experiments were conducted in a controlled environment involving both host and guest operating systems, as well as various simulation and network analysis tools. The host system, which ran Windows 11 on an Intel Core i5 processor, provided the required computational resource. The guest operating system, Ubuntu 24.04.2 LTS, was selected for its stability and suitability for the network simulations. Virtualisation was managed using VirtualBox 7.1.6, which enabled the creation and management of virtual machines for testing purposes. The network topology and simulations were emulated using Mininet, a tool commonly employed to simulate an SDN. The network was controlled by a single Ryu controller, with the number of switches and hosts determined by the topology. Traffic generation and monitoring were performed using mgen and tcpdump, respectively, to simulate and capture the network activity. Communication between network devices utilized the OpenFlow protocol, with port 6653 configured for connection. The network topologies were configured and visualized using the MiniEdit package. The simulations lasted for 280 s and involved six different topologies. ML methods were developed and executed using Python 3.13.3 within an Anaconda environment, ensuring a well-managed and isolated setup. TensorFlow was used as the primary deep learning framework for model implementation and training.

It should be noted that while Mininet provides an effective and flexible environment for functional validation and prototyping, it does not fully capture the throughput and scalability characteristics of carrier-grade SDN deployments. The traffic rates achievable in Mininet are inherently limited by the host machine's resources and cannot emulate multi-Gbps conditions. Therefore, the results reported in this study should be interpreted as feasibility validation under emulated conditions, with large-scale throughput evaluations reserved for future work on hardware-based SDN testbeds.

H. Performance Evaluation Metrics

Evaluating the performance of AI techniques is a critical step in establishing their effectiveness, especially in sensitive domains such as SDN security [1], [19]. In the context of DDoS attack detection, an accurate assessment of method behavior ensures that the solution not only detects intrusions reliably but also minimizes the risk of false alarms that may

TABLE V
CONFUSION MATRIX FOR BINARY CLASSIFICATION

Population	Predicted Condition	
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

disrupt legitimate traffic. Therefore, statistical performance metrics such as “accuracy”, “precision”, “recall”, and the “F1-score” are indispensable for quantifying framework reliability and robustness. The evaluation process is based on the confusion matrix, which is a tabular representation of the classification outcomes. Table V presents the confusion matrix for binary classification, where the approach distinguishes between benign and malicious traffic (i.e., DDoS traffic).

Each term in the confusion matrix serves a distinct evaluative role.

- **TP:** Number of malicious flows correctly identified as attacks.
- **TN:** Number of benign flows correctly classified as non-attacks.
- **FP:** Benign flows are incorrectly flagged as attacks, contributing to false alarms.
- **FN:** Malicious flows are misclassified as benign, indicating a failure in threat detection.

These values are subsequently used to compute the core performance metrics, as defined below.

$$\text{Accuracy (ACC)} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$\text{Precision (PRC)} = \frac{TP}{TP + FP} \quad (4)$$

$$\text{Recall (RCL)} = \frac{TP}{TP + FN} \quad (5)$$

$$\text{F1 Score} = 2 \cdot \frac{\text{RCL} \cdot \text{PRC}}{\text{RCL} + \text{PRC}} \quad (6)$$

- The ACC provides a global measure of correctness but may be misleading in imbalanced datasets, where attack instances are rare.
- The PRC emphasizes the reliability of positive predictions and is crucial for minimizing the cost of false alarms in critical SDN environments.
- The RCL (also known as sensitivity or true positive rate) indicates the model’s effectiveness in identifying all instances of attacks, thus measuring detection coverage.
- The F1-score offers a harmonic mean of precision and recall, making it a balanced metric when the dataset exhibits class imbalance or when both false positives and false negatives carry significant consequences.

IV. RESULTS AND DISCUSSION

This section presents an evaluation of various ML approaches, focusing on their precision in detecting SYN-based DDoS attacks. The assessment considers several effectiveness parameters that, although not fully detailed here, form the basis of the analysis. The evaluation exclusively used SYN-type DDoS attack data. The dataset was divided into

TABLE VI
PERFORMANCE OF PARAMETERS RESULTS

Technique	Accuracy	Precision	Recall	F1-Score
DT	96.8%	95.3%	94.7%	94.9%
RF	97.1%	93.0%	95.0%	93.9%
CNN	98.2%	96.3%	96.5%	96.4%
LSTM	95.0%	94.4%	93.8%	94.1%
CNN-LSTM	99.5%	98.34%	97.2%	97.7%

two parts: 80% was used for training and 20% for testing, ensuring a rigorous assessment of the system’s performance. The total size of the dataset for this evaluation was 5.12 GB, providing ample data to assess the accuracy and reliability of the detection process across different scenarios.

Furthermore, this study evaluated the performance of several ML methods, as listed in Table VI. The CNN-LSTM hybrid approach outperformed the other approaches, achieving the highest F1-Score of 97.77%, thereby demonstrating its effectiveness in combining convolutional and sequential learning approaches. The CNN technique closely followed, with an F1-Score of 96.40%, reflecting its solid precision and recall performance. Both the DT and RF frameworks also showed competitive results, with F1-Scores of 94.99% and 93.99%, respectively. However, the balance between precision and recall was not as optimized as that of the deep learning models. Although the LSTM model was effective, it achieved a lower F1-Score of 94.10%, suggesting that the CNN-based system provided a stronger performance for this custom dataset. The results, as outlined in Table VI, emphasize the superiority of hybrid techniques, such as CNN-LSTM, in boosting classification accuracy and showcase the significant advantages of DL architectures in handling complex tasks that demand high precision and reliability.

A. Redirecting Network Traffic for DDoS Mitigation

To mitigate the disruptive impact of DDoS attacks, the proposed strategy employs intelligent traffic redirection as a proactive defence mechanism. The core concept involves discreetly rerouting malicious traffic to alternate destinations, such as honeypots, intrusion prevention systems, or traffic analysis platforms, without alerting the attacker. This ensures the continuity of legitimate services while allowing network security teams to monitor and respond to evolving threats effectively. A key element of this architecture is the redirection of malicious DDoS traffic to a cloud-based honeypot system. Once traffic is flagged as suspicious, it is rerouted from the primary network infrastructure to the honeypot, where a detailed behavioral analysis can be conducted in isolation. This controlled environment provides valuable insights into attack patterns, techniques, and potential vulnerabilities that can be exploited without compromising critical services.

The detection and rerouting processes are triggered at the network edge, where an edge device acts as a redirect server. Upon detecting DDoS indicators, the edge device uses an edge switch to divert harmful traffic away from the main data path, thereby significantly reducing the resource strain on the core

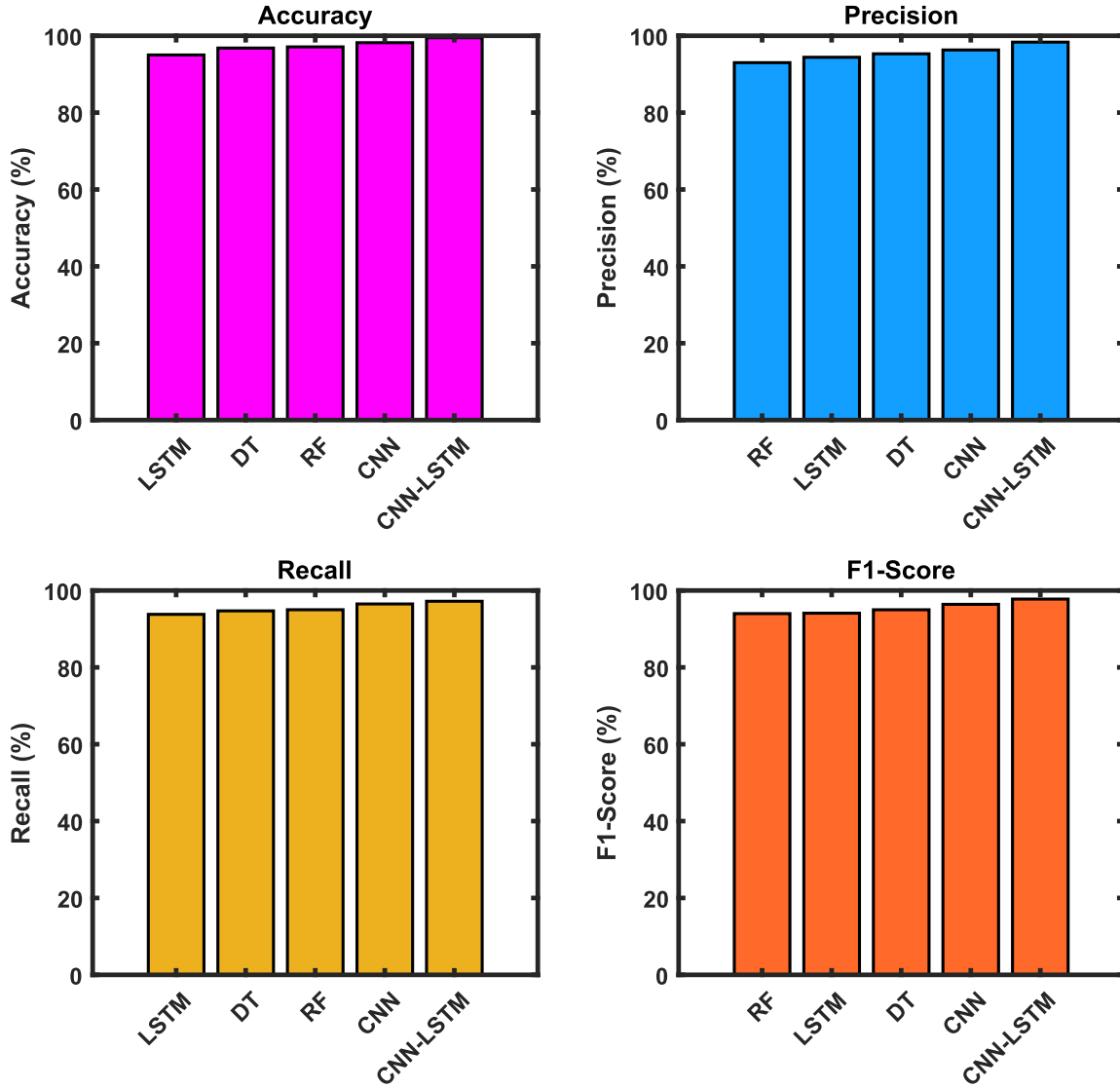


Fig. 8. Comparative performance analysis of classical ML algorithms (Decision Tree, Random Forest) and DL architectures (LSTM, CNN, CNN-LSTM) for DDoS traffic classification. Each model is evaluated using Accuracy, Precision, Recall, and F1-Score. The hybrid method consistently delivers high accuracy, making it ideal for intrusion detection in SDN environments.

network. Simultaneously, legitimate traffic is forwarded to its intended destination to ensure minimal disruption to users. Figure 7 illustrates the DDoS traffic filtering and redirection flow, showing the decision-making process from the initial detection to the active response. The flowchart highlights how traffic is filtered and either blocked or redirected, depending on its classification. The performance metrics validate the effectiveness of this strategy. After rerouting, the network experienced a notable decrease in both bandwidth consumption and packet rates, confirming a substantial reduction in the attack load. This shift in traffic patterns underscores the efficiency of redirection as the core DDoS mitigation approach.

To prevent any possibility of data leakage, the split was performed at the flow level, ensuring that no identical or overlapping flows appeared in both training and testing sets. Furthermore, the model was validated across six distinct SDN topologies, which provided additional confirmation of

its ability to generalize beyond a single network configuration. The learning curves presented in Figures 10 and 11 illustrate close alignment between training and testing accuracy and loss, which indicates that the model maintained generalization capability without signs of overfitting. In addition to accuracy, balanced precision (98.34%), recall (97.2%), and F1-score (97.7%) further support the robustness of the framework.

B. Discussion

Figure 8 illustrates the comparative performance of the five classification techniques of DT, RF, CNN, LSTM, and the proposed CNN-LSTM hybrid evaluated on a custom SDN dataset. These approaches were assessed using four key metrics: accuracy, precision, recall, and F1-score. CNN-LSTM achieved the highest accuracy of 99.5%, outperforming CNN (98.2%) and RF (97.1%). DT and LSTM followed with slightly lower scores of 96.8% and 95.0%, respectively. This highlights

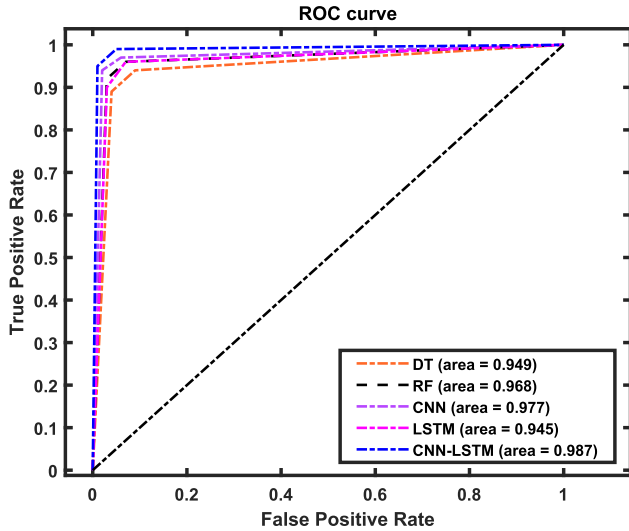


Fig. 9. ROC curves illustrating the classification performance.

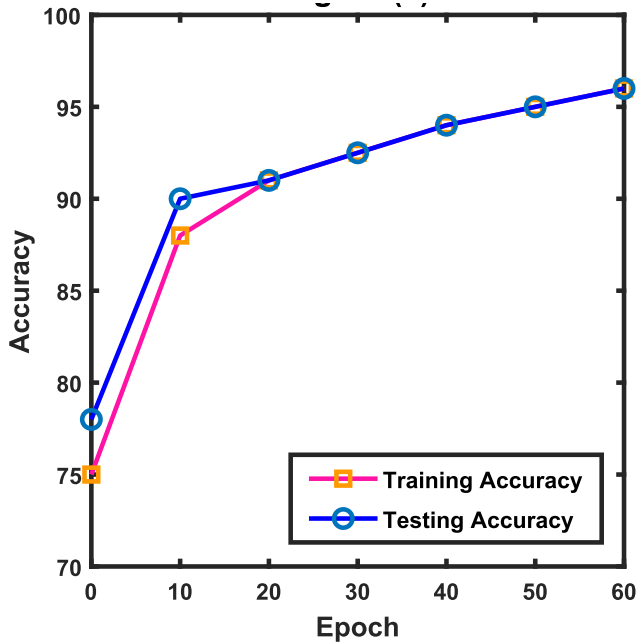


Fig. 10. Accuracy: Training vs Testing.

the effectiveness of CNN-LSTM in accurately distinguishing between normal and malicious SDN traffic. The evaluation of precision, recall, and F1-score, as presented in Figure 8, further emphasizes the dominance of CNN-LSTM. With a precision of 98.34% and a recall of 97.2%, it demonstrates strong reliability in minimizing both false positives and false negatives. Its F1-score of 97.77% confirms a well-balanced performance. Although CNN, RF, and DT yielded competitive results (ranging from 93% to 96%), they exhibited slightly higher misclassification rates than the other models. Although LSTM was still effective, it showed the lowest precision (94.4%), recall (93.8%), and F1-score (94.10%), indicating reduced consistency in identifying attack patterns. Compared with our previous work, which employed individual machine

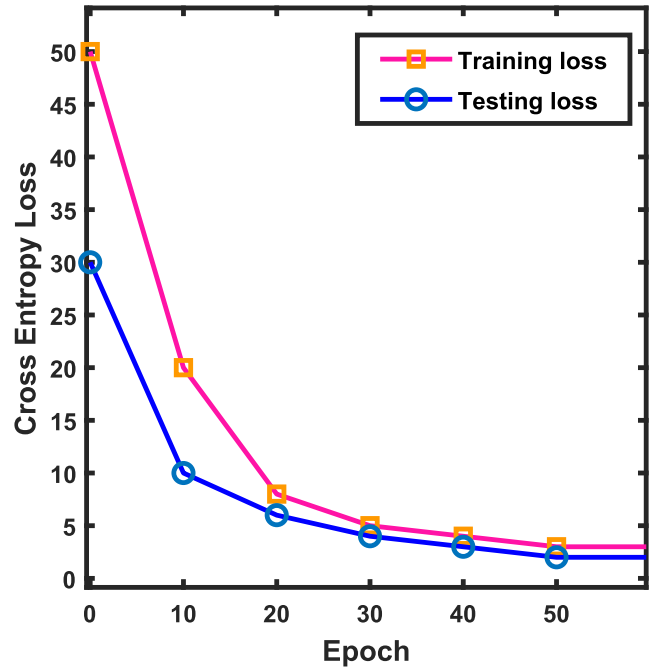


Fig. 11. Loss: Training vs Testing.

learning algorithms for DDoS detection [73], this hybrid DL approach demonstrates significantly improved performance and offers enhanced security for SDN environments.

The Receiver Operating Characteristic (ROC) curve serves as a robust and threshold-independent evaluation metric, widely recognized for its effectiveness in assessing the discriminative ability of classification models. By illustrating the trade-off between the true positive and false positive rates across varying decision thresholds, it provides a comprehensive view of the model performance beyond fixed cut-off values. As depicted in Figure 9, the comparative analysis revealed distinct ROC curve trajectories, with the corresponding Area Under the Curve (AUC) values offering quantifiable insights into classification efficacy. Specifically, the DT, RF, CNN, LSTM, and CNN-LSTM models achieved AUC values of 0.949, 0.968, 0.977, 0.945, and 0.987, respectively. Among these, the CNN-LSTM attained the highest AUC, confirming its superior predictive capability, while the remaining models demonstrated competitive results.

Figure 10 and 11 illustrates the learning behavior of the CNN-LSTM architecture across 60 training epochs, tracking the progression of accuracy and cross-entropy loss, respectively. In Figure 10, both the training and testing accuracies steadily increase, converging at approximately 96%. The close alignment of these curves indicates a strong generalization with minimal signs of overfitting. Figure 11 shows a sharp reduction in the loss during the initial epochs, followed by a gradual stabilization near low values (approximately 2 to 3). This trend reflects efficient parameter optimization and a smooth convergence process that is free from instability or degradation. Taken together with the performance results from Figure 10 and 11, these insights affirm the robustness of

TABLE VII

PERFORMANCE COMPARISON OF RECENT DDoS DETECTION AND ATTACK RESPONSE APPROACHES IN SDN. **M**:: MITIGATION, **D**:: DETECTION

Reference	Techniques	ACC	PRC	RCL	F1-Score	Dataset	M	D
<i>Singh et al. [39]</i>	SVM+RF	99.1%	99.1%	98.8%	99.2%	Self-Generated	✓	✓
<i>Wldhai et al. [41]</i>	ARF+KNN	95.0%	87.0%	87.0%	87.0%		✗	✓
<i>Manisankar et al. [43]</i>	RF+GB	97.0%	96.96%	98.0%	97.4%		✓	✓
<i>Tonkal et al. [44]</i>	ML+NCA	99.1%	99.1%	98.1%	99.3%		✓	✓
<i>Sangodoyin et al. [46]</i>	ML+CFA	98.9%	-	-	-		✗	✓
<i>Gadallah et al. [47]</i>	AE+BGRU	97.1%	98.9%	99.4%	99.1%		✓	✓
<i>Said et al. [49]</i>	CNN + BiLSTM	97.7%	99.8%	95.2%	97.5%		✓	✓
<i>Zhang & Wang [52]</i>	SSAE+SVM	98.6%	-	-	-		✓	✓
<i>Mhamdi & Isa [56]</i>	AE+RF	98.0%	98.8%	98.9%	98.9%		✓	✓
<i>Pradeesh et al. [57]</i>	AEL+ Ensemble	96.7%	96.7%	96.7%	96.7%		✓	✓
Our proposed	CNN-LSTM	99.5%	98.34%	97.2%	97.7%		✓	✓

CNN-LSTM in achieving high classification accuracy and maintaining reliable and consistent learning dynamics.

In addition to accuracy-based metrics, we monitored the system resource utilization when deploying the CNN-LSTM model inside the Ryu controller. During inference, the model introduced a modest computational overhead, with CPU usage remaining below 12% and memory consumption below 9% of the available system resources. These results indicate that the framework can operate efficiently within the constraints of the SDN controller environment. Furthermore, the reduction of input dimensionality to 16 features through feature selection significantly contributed to lowering the inference cost without sacrificing detection accuracy.

While the technique demonstrated efficient detection and mitigation in the Mininet-based environment, we acknowledge that this platform cannot emulate the full traffic intensity of real-world multi-Gbps networks. Consequently, although our results confirm instantaneous responsiveness under emulation, large-scale validation under production-level throughput remains a subject for future research.

C. Comparison to Other Similar Methods

Table VII provides a comparative summary of our proposed CNN-LSTM framework against several state-of-the-art DDoS detection and mitigation techniques in SDN environments. The comparison is structured around standard classification metrics, namely ACC, PRC, RCL, and F1-score, as well as the dataset origin and functional scope of each study, that is, detection (D) and mitigation (M). Our proposed model consistently demonstrates superior overall performance, achieving 99.5% accuracy, 98.34% precision, 97.2% recall, and 97.7% F1-score, outperforming all other approaches in the table. While prior algorithms, such as those by Singh et al. [39], Tonkal et al. [44], and Gadallah et al. [47] reported high individual metrics (some exceeding 99% in accuracy or recall), most of which failed to achieve a balanced performance across all evaluation dimensions, particularly when paired with effective defense mechanism. A critical observation from this comparison is that many existing studies either lack traffic control strategy capabilities (for example, Eldhai et al. [41], Gadallah et al. [47], and Tonkal et al. [44]) or rely on

traditional ML approaches that may struggle with complex or evolving attack vectors. In contrast, our framework leverages the complementary strengths of CNN for spatial feature extraction and LSTM for temporal behavior modeling, resulting in a more adaptive and context-aware detection pipeline. Furthermore, our integrated script-based mitigation within the SDN controller enables an immediate traffic response, an aspect that is underdeveloped or absent in many prior studies.

Dataset transparency and availability are equally important. While several studies have utilized self-generated datasets, such as Singh et al. [39] Gadallah et al. [47], Said et al. [49], and Mhamdi and Isa [56] have not publicly released their datasets, limiting reproducibility and independent benchmarking. In contrast, we plan to publicly release our custom SDN-based dataset, which includes realistic SYN and DoS traffic captured in a Mininet Ryu testbed. This contribution not only supports future benchmarking efforts but also promotes open research and standardization in the SDN security community. Finally, the end-to-end design of our framework, which encompasses data collection, hybrid deep learning, and mitigation, distinguishes it from other frameworks that often address only isolated components of the problem. As such, our approach is not only technically superior in terms of classification metrics, but is also more deployable, transparent, and scalable, offering a meaningful advancement in the field of intelligent DDoS defense for SDN.

Importantly, these findings tie directly to the research gaps identified in Section II. First, unlike many prior works that relied on outdated or non-public datasets, our custom SDN dataset enhances reproducibility and realism. Second, we applied a principled multi-stage feature selection pipeline to reduce redundancy and computational cost, a step often overlooked in earlier studies. Third, our hybrid CNN-LSTM model addresses the clear need for architectures that capture both spatial and temporal flow characteristics, outperforming standalone ML/DL baselines. Finally, by embedding attack response logic directly within the SDN controller, our framework closes the common gap of detection-only solutions, enabling instantaneous responsiveness with minimal overhead. This explicit mapping of results to unresolved challenges reinforces the novelty and deployment feasibility of our approach.

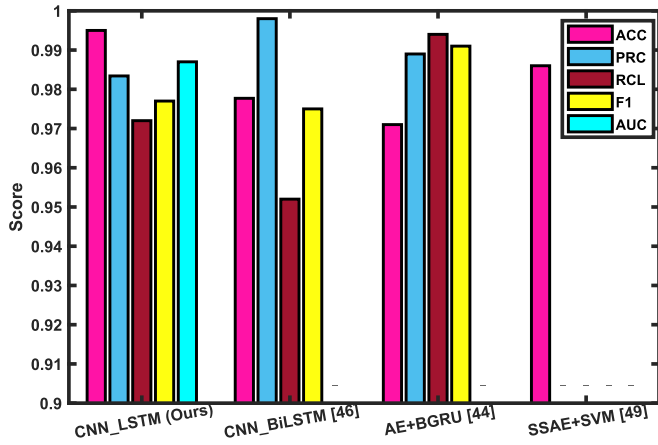


Fig. 12. Comparison with recent DL/hybrid approaches (CNN-BiLSTM, AE+BGRU, SSAE-SVM) on key metrics.

To further broaden the comparison, we also benchmarked our CNN-LSTM model against recent deep and hybrid architectures such as CNN-BiLSTM, AE+BGRU, and SSAE-SVM. The results, summarized in Figure 12, show that our framework consistently achieved superior F1-score (97.7%) and AUC (0.987), while maintaining competitive precision and recall. Although CNN-BiLSTM and AE+BGRU achieved comparable accuracy, their performance balance across all metrics was less stable. SSAE-SVM demonstrated strong accuracy but exhibited a slightly higher false positive tendency. These results highlight that the proposed CNN-LSTM not only achieves state-of-the-art detection accuracy, but also ensures robustness across multiple evaluation dimensions, directly addressing recent calls for stronger DL/hybrid baselines in SDN DDoS detection.

V. CONCLUSION

This study presents a hybrid deep learning framework based on a CNN-LSTM architecture for detecting and mitigating DDoS attacks in SDN environments. The proposed approach leverages a custom-built dataset that reflects realistic SDN traffic characteristics, ensuring high detection reliability and reproducibility. Through comprehensive experiments, the CNN-LSTM technique achieved a detection accuracy of 99.5% and an F1-score of 97.7%, outperforming traditional ML models and standalone DL techniques. In addition to detection, the system integrates a script-based mitigation mechanism within the SDN controller, allowing for identification and blocking or redirection of malicious traffic. This enables a fast response without relying on third-party tools, ensuring lightweight deployment and scalability across various network topologies. Our results demonstrate the effectiveness of combining spatial and temporal learning for intrusion detection and highlight the value of integrating adaptive, controller-level mitigation.

Future research will focus on expanding the system to support multi-class attack classification and incorporating transformer-based architectures and reinforcement learning to enhance adaptability against evolving threats. Furthermore,

we recognize that the present dataset primarily includes network- and transport-layer DDoS scenarios (SYN, UDP, ICMP floods), while low-rate and application-layer attacks such as HTTP floods or slowloris were not yet incorporated. We plan to extend the dataset with these attack categories and validate our framework on large public benchmarks (e.g., CICDDoS2019, CICIOT2023) to ensure robustness against a broader spectrum of threat vectors. Furthermore, we plan to enhance the mitigation component by extending beyond script-based packet dropping and rerouting. Future work will explore adaptive and policy driven approaches, such as QoS-aware throttling, traffic shaping, and reinforcement learning assisted flow management, to provide more granular and scalable responses to diverse attack scenarios. In addition, future research will investigate model optimization techniques such as pruning, quantization, and the adoption of lightweight deep learning architectures (e.g., MobileNet or GRU-based variants) to further reduce computational overhead and ensure scalability in production-grade SDN environments.

DATA AVAILABILITY

The dataset and related supplementary materials are not publicly available but can be accessed upon request. A public GitHub repository has been created to manage such requests and provide documentation (<https://github.com/engrabbinasir/hybrid-cnn-lstm-ddos-sdn>). Interested researchers may contact us with their name, institution, and intended use.

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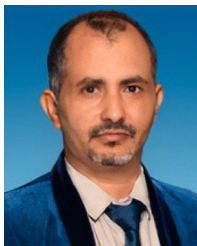
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